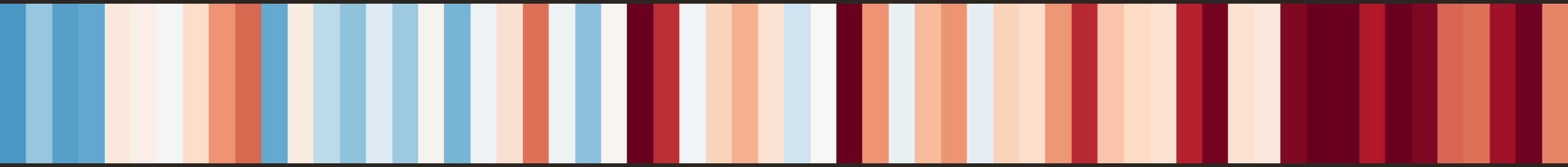
DATA ANALYSIS PROJECT

Is South Carolina Heating Up?

Three angles on 60 years of NOAA weather data (1966-2025)

Ankit Saini | Artificial Intelligence | Dr. Songhui Yue | November 1, 2026



1966 2025

WHY THIS TOPIC

Why South Carolina?

What has actually changed in South Carolina's weather over the last 60 years: **where, when, and how do people feel it?**

I used public NOAA data that anyone can download, and tested the changes with statistics instead of just looking at charts.

1

Floods

Charleston floods at high tide, and the state had a "1000-year" flood in October 2015.

2

Heat

Summers are hot and humid. Hot nights affect health and electricity bills.

3

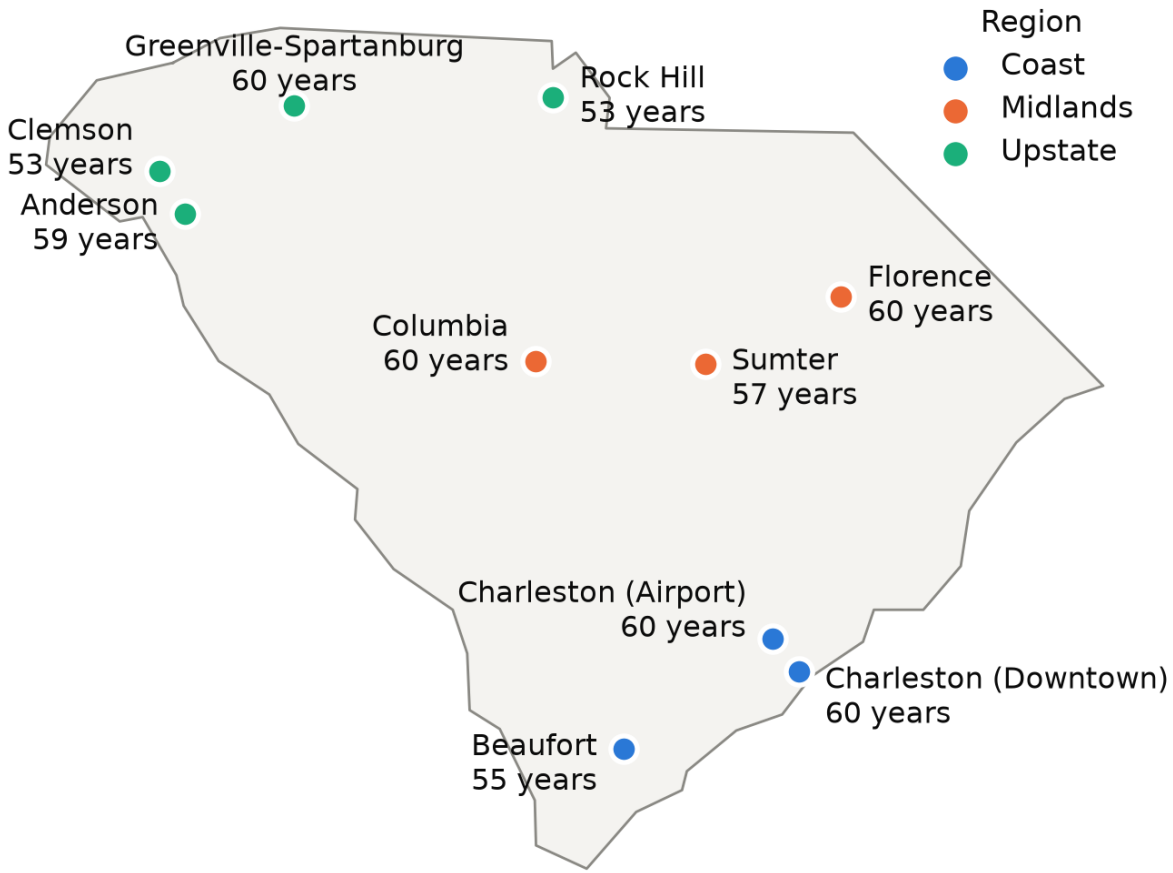
Farms

Agriculture depends on freeze dates and the length of the growing season.

Source: NOAA NCEI GHCN-Daily, 10 South Carolina stations, 1966-2025

TASK 1: THE DATA

The data: 10 NOAA weather stations, 1966-2025



10
stations in 3 regions

60
years: 1966-2025

217,782
days of highs, lows and rain

Source: NOAA NCEI GHCN-Daily (the data behind NOAA's "Past Weather" page)

Download: one CSV file per station from a public URL, no account needed

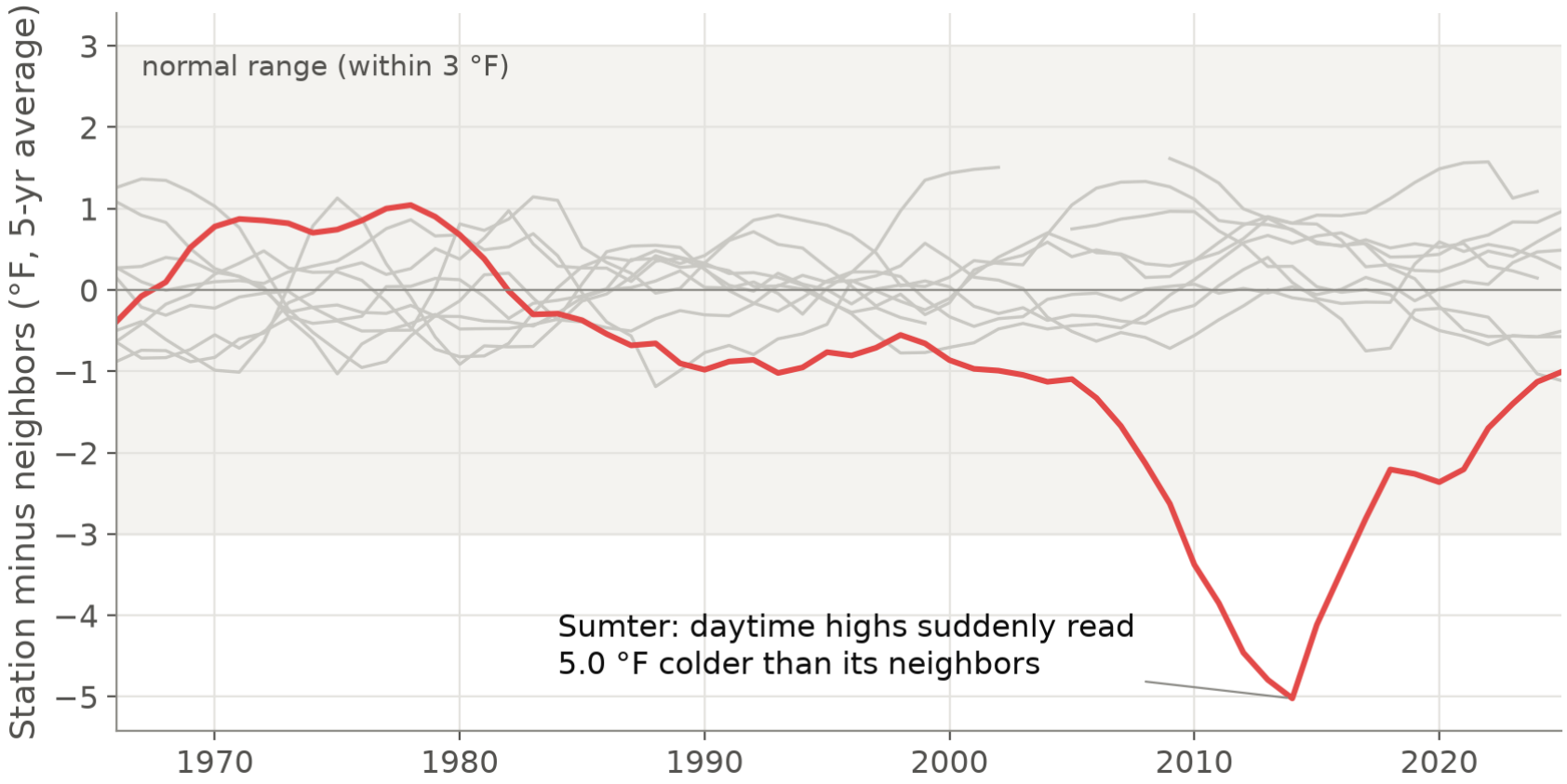
Selection: 18 candidates → kept 10 with ≥ 330 days of data in (almost) every year

Variables: daily high (TMAX), daily low (TMIN), rain (PRCP)

Source: NOAA NCEI GHCN-Daily, 10 South Carolina stations, 1966-2025

TASK 2: CLEANING

Cleaning and checking the data



−5 °F

Sumter's error compared with nearby stations (2010-2015), so I left it out of statewide averages

- Converted to °F and inches
- Removed values that failed NOAA's quality flags
- Kept only years with ≥ 330 days
- Neighbor check: gray lines are normal stations

Source: NOAA NCEI GHCN-Daily, 10 South Carolina stations, 1966-2025

THE THREE ANGLES

Three angles

1 WHERE

Which city has warmed the most?

Tells us if warming is local or statewide, and ranks cities for local planning. (The assignment's example question.)

2 WHEN

Day or night? Which season?

The yearly average hides how it warms. Night vs. day points to causes like city heat; seasons matter for crops and energy.

3 WHAT PEOPLE FEEL

Are extreme days more common?

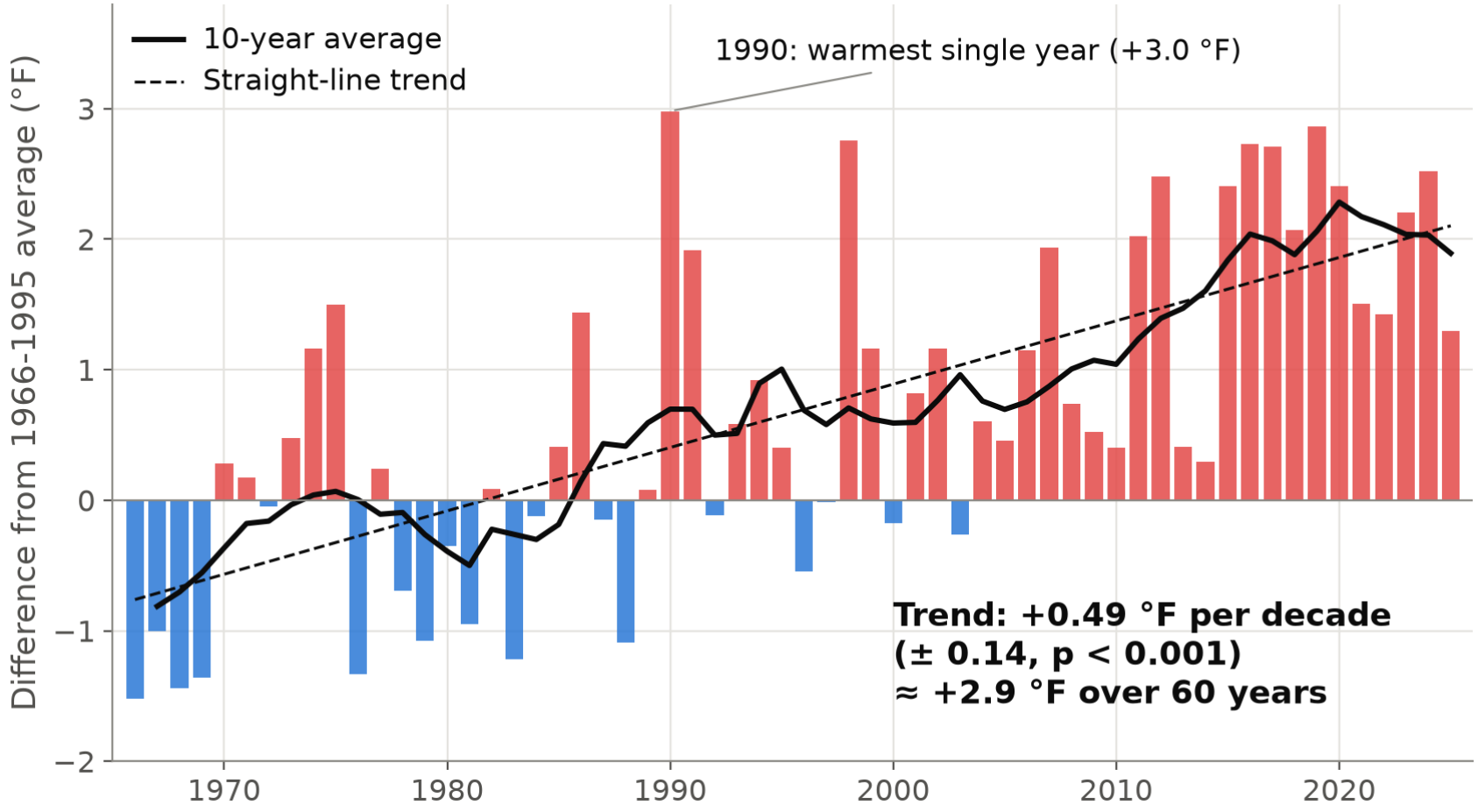
People notice thresholds: a 95 °F afternoon, a night that never cools off, a frost, a flood. This also tests rainfall.

the average → **its timing** → **the extremes** (all three use the same data, a straight-line trend with a p-value, and 1966-1995 vs. 1996-2025)

Source: NOAA NCEI GHCN-Daily, 10 South Carolina stations, 1966-2025

ANGLE 1: WHERE

South Carolina has warmed about 3 °F since the 1960s



+2.9 °F

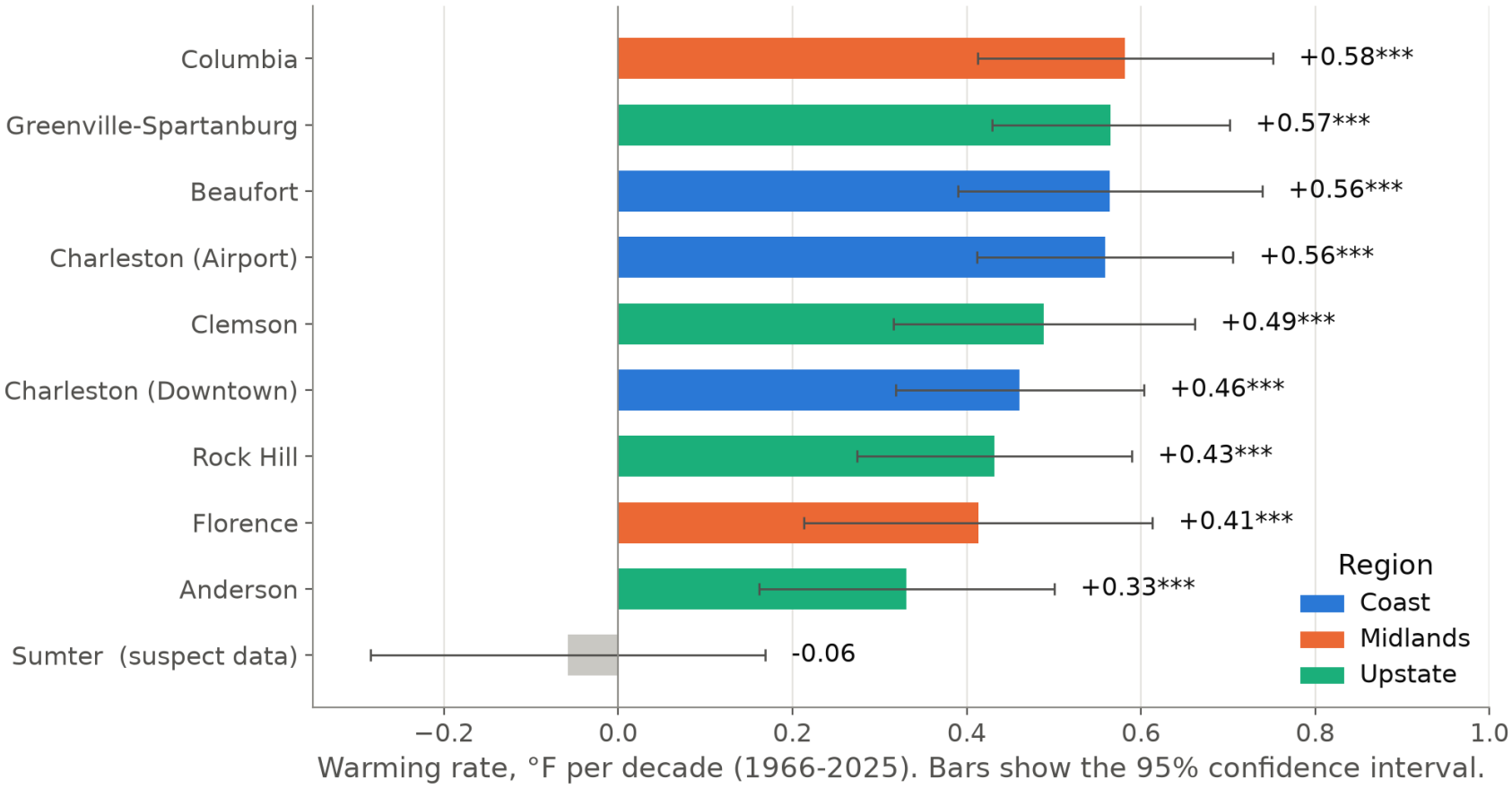
over 60 years (+0.49 °F per decade, $p < 0.001$)

- Before 1985 most years are blue; since 2010 almost all are red
- 8 of the 10 warmest years came after 2010
- Average of the 9 trusted stations

Source: NOAA NCEI GHCN-Daily, 10 South Carolina stations, 1966-2025

ANGLE 1: WHERE

Columbia warmed fastest, but the top four are close



+0.58

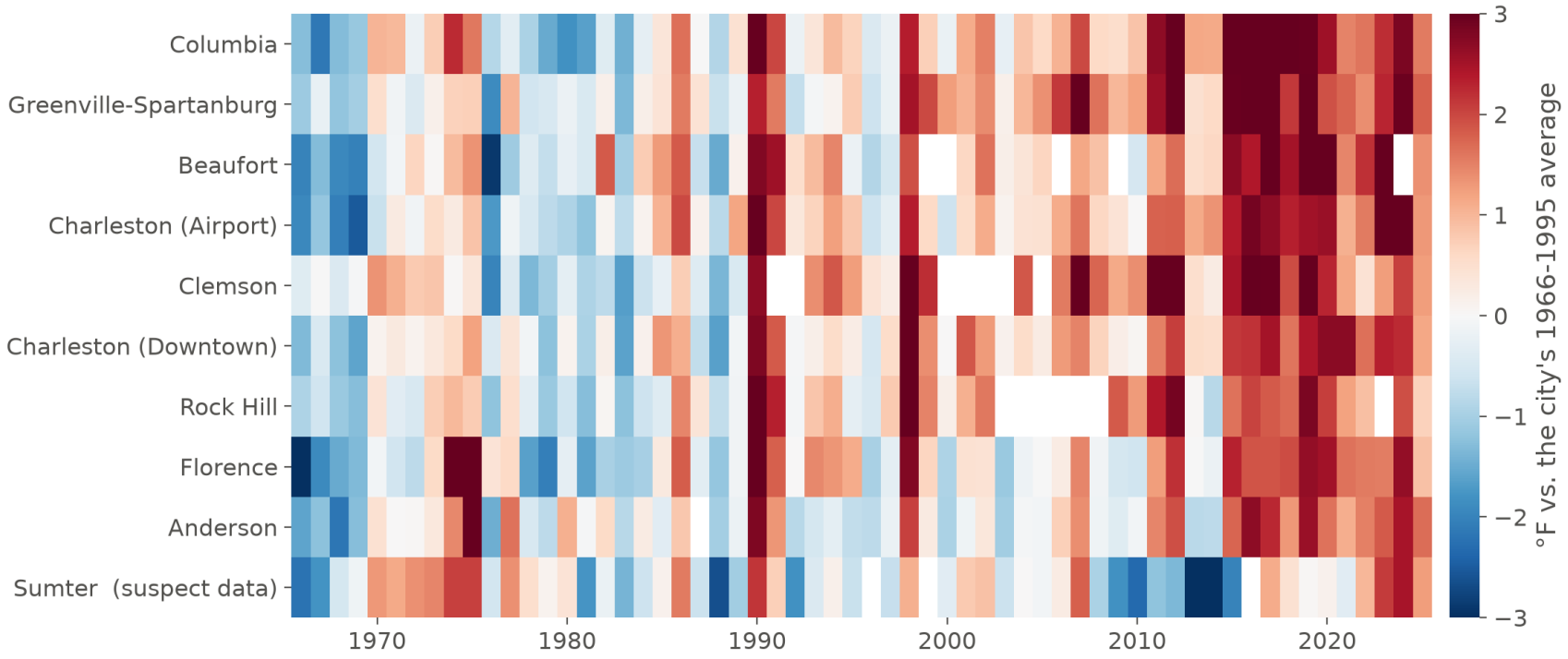
°F per decade in Columbia, about +3.5 °F since 1966

- Top 4 are within 0.03 °F/decade and their error bars overlap
- All 9 trusted stations warmed ($p < 0.001$)
- Coast 0.53, Midlands 0.50, Upstate 0.45

Source: NOAA NCEI GHCN-Daily, 10 South Carolina stations, 1966-2025

ANGLE 1: WHERE

Every station shows the same pattern



9 / 9

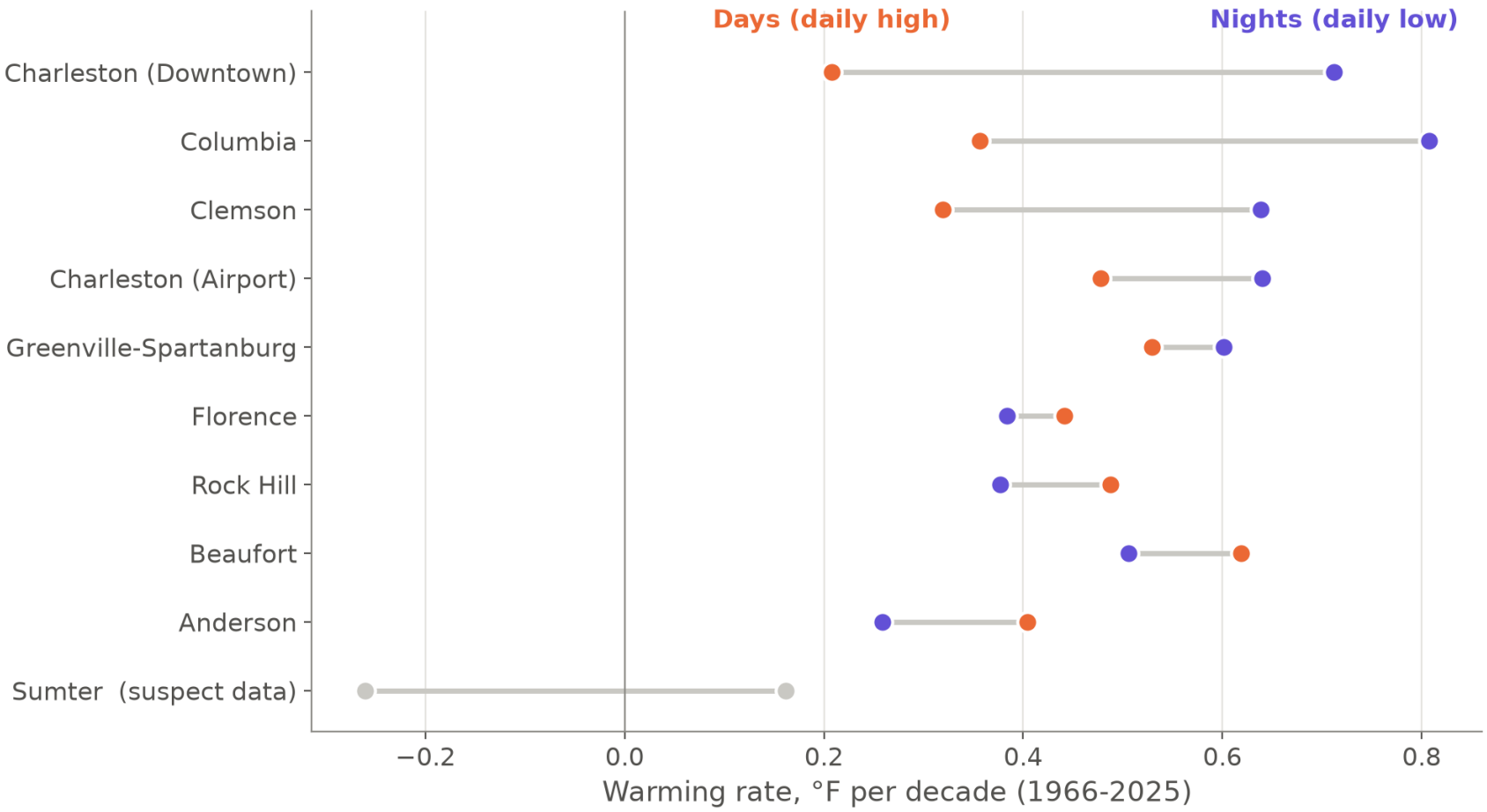
trusted stations have a significant warming trend

- Each row = a city, each column = a year
- Red = warmer than that city's own 1966-95 normal
- Sumter's broken years show as dark blue (2013-14)

Source: NOAA NCEI GHCN-Daily, 10 South Carolina stations, 1966-2025

ANGLE 2: WHEN

Nights are warming faster than days



3×

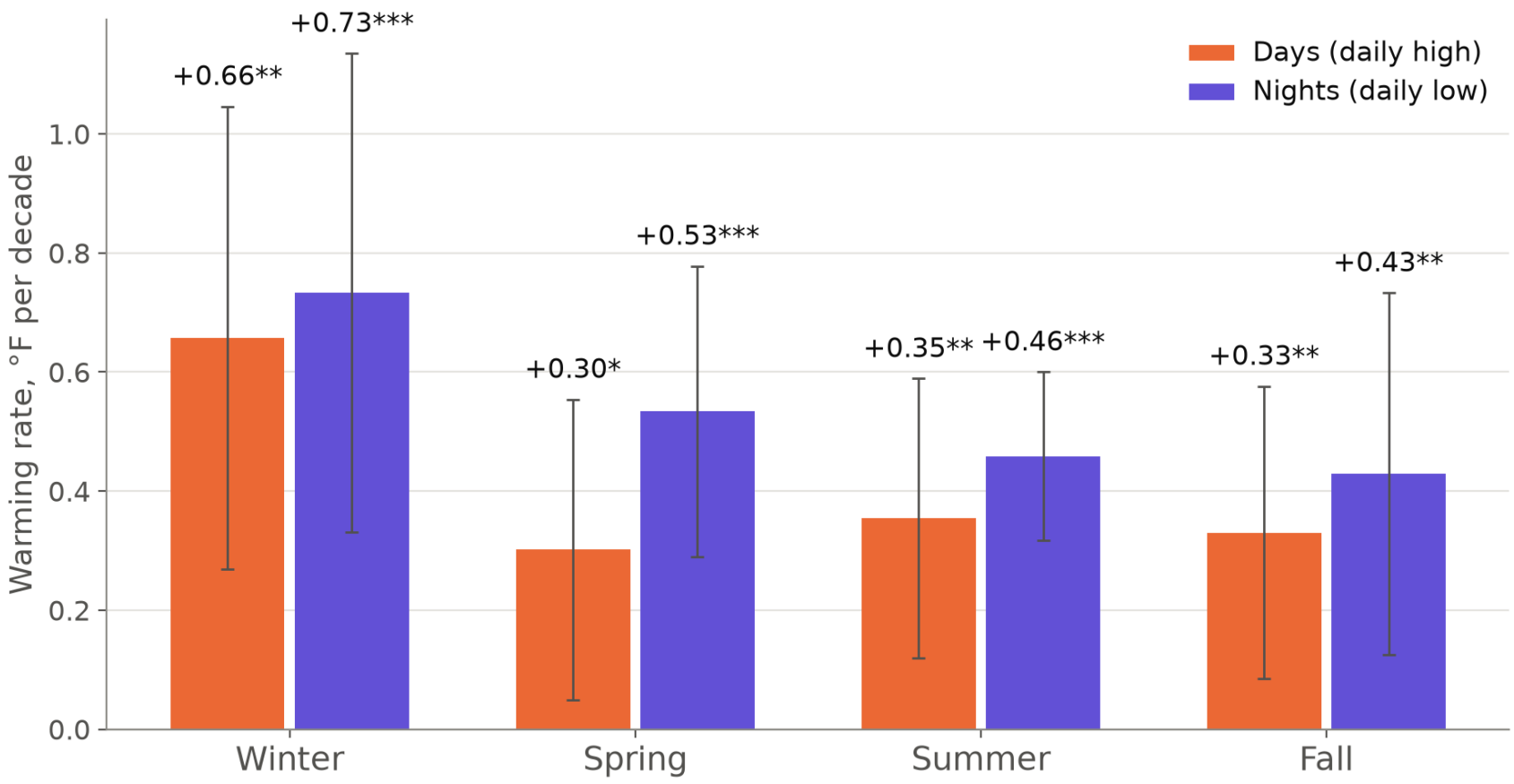
faster warming at night than by day in downtown Charleston

- Average: nights +0.55 vs. days +0.43 °F/decade
- Biggest gaps: downtown Charleston and Columbia
- Urban heat island: concrete stores heat, releases it at night

Source: NOAA NCEI GHCN-Daily, 10 South Carolina stations, 1966-2025

ANGLE 2: WHEN

Winter is warming fastest



1.7×

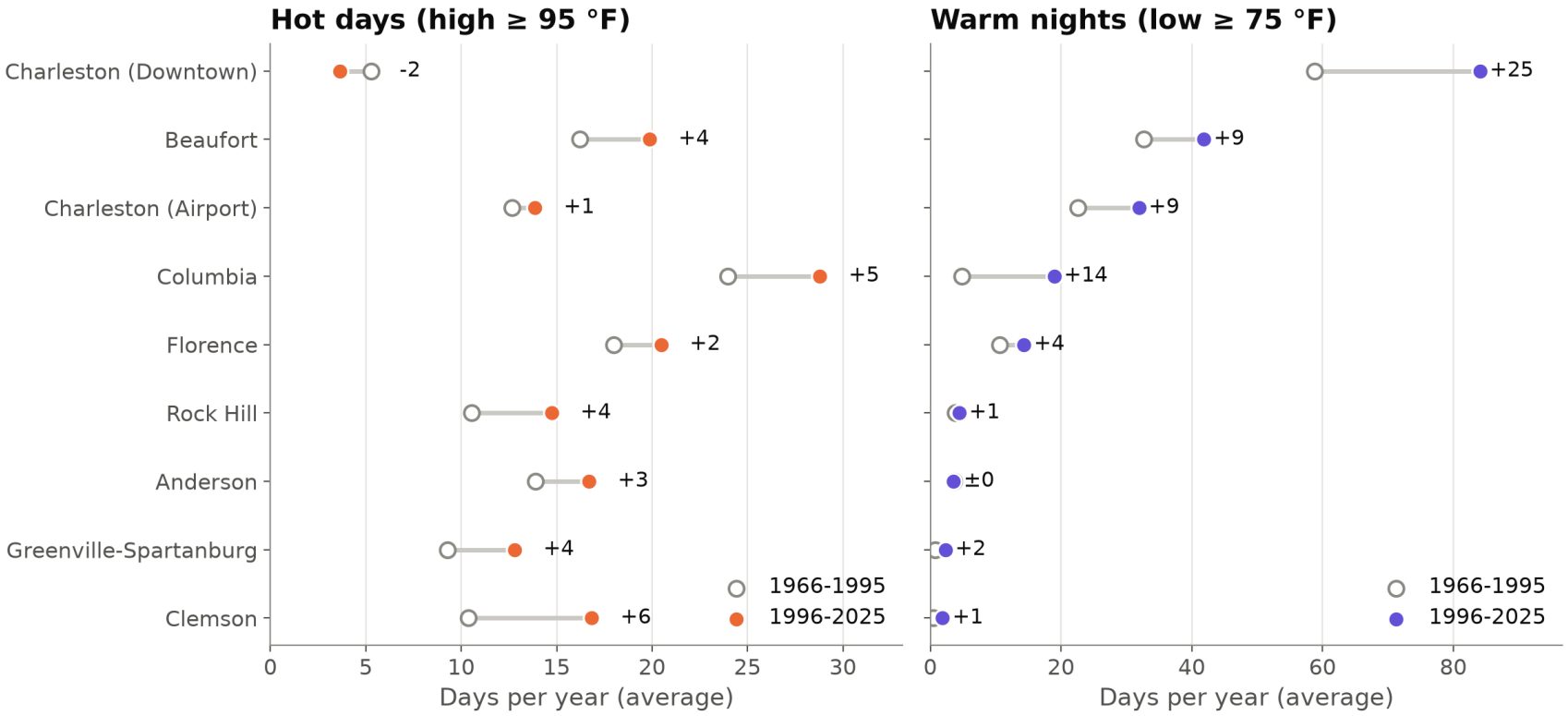
winter warms 1.7 times faster than the other seasons

- Winter: +0.66 (days), +0.73 (nights) °F/decade
- Other seasons: nights warm faster than days
- Winter varies a lot year to year → wide error bars

Source: NOAA NCEI GHCN-Daily, 10 South Carolina stations, 1966-2025

ANGLE 3: WHAT PEOPLE FEEL

Many more warm nights



+47%

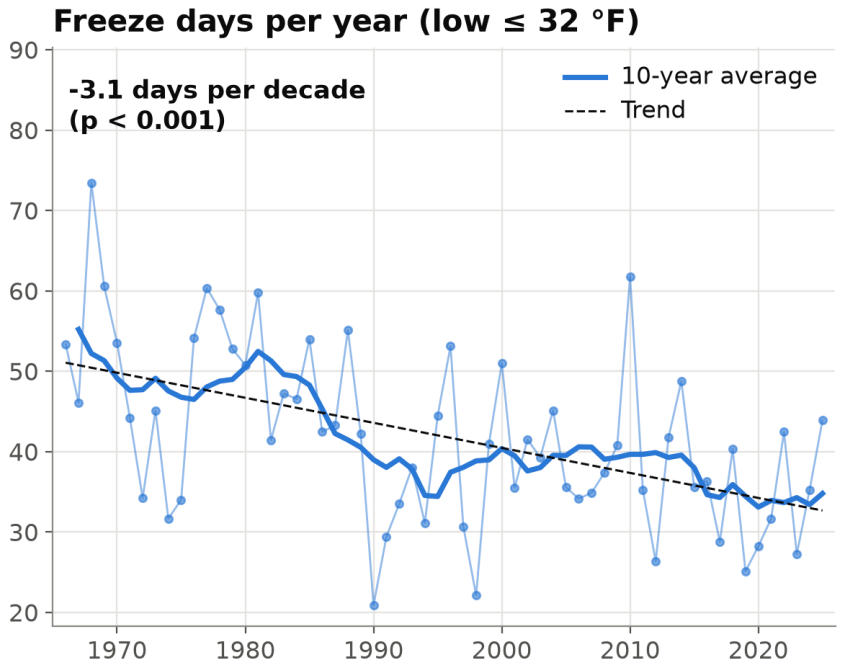
more warm nights statewide (15.6 → 22.9 per year)

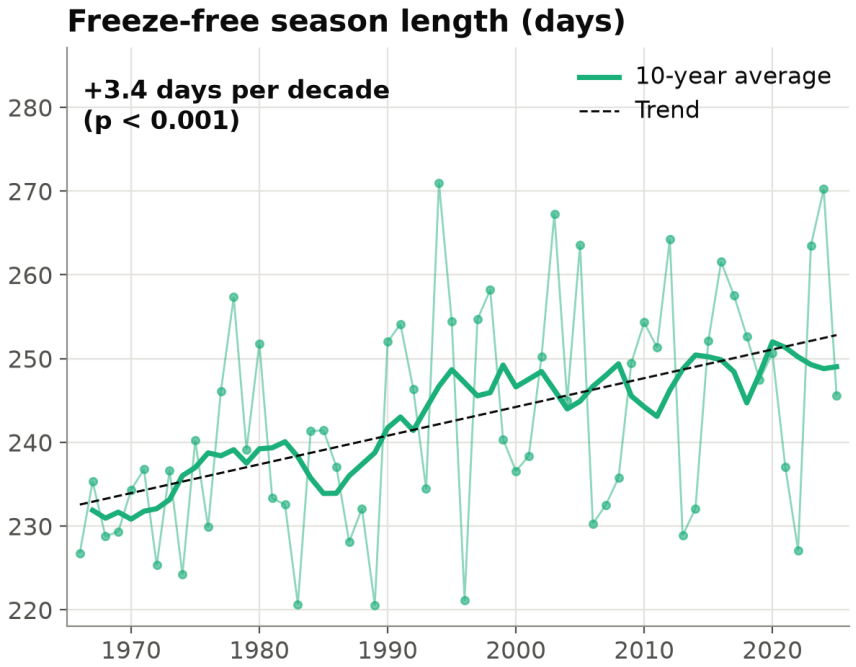
- Downtown Charleston: 59 → 84 warm nights a year
- Columbia: 5 → 19 (nearly 4×)
- Hot days ≥ 95 °F: +21% (just significant)

Source: NOAA NCEI GHCN-Daily, 10 South Carolina stations, 1966-2025

ANGLE 3: WHAT PEOPLE FEEL

Fewer freezes and a longer growing season





+3 weeks

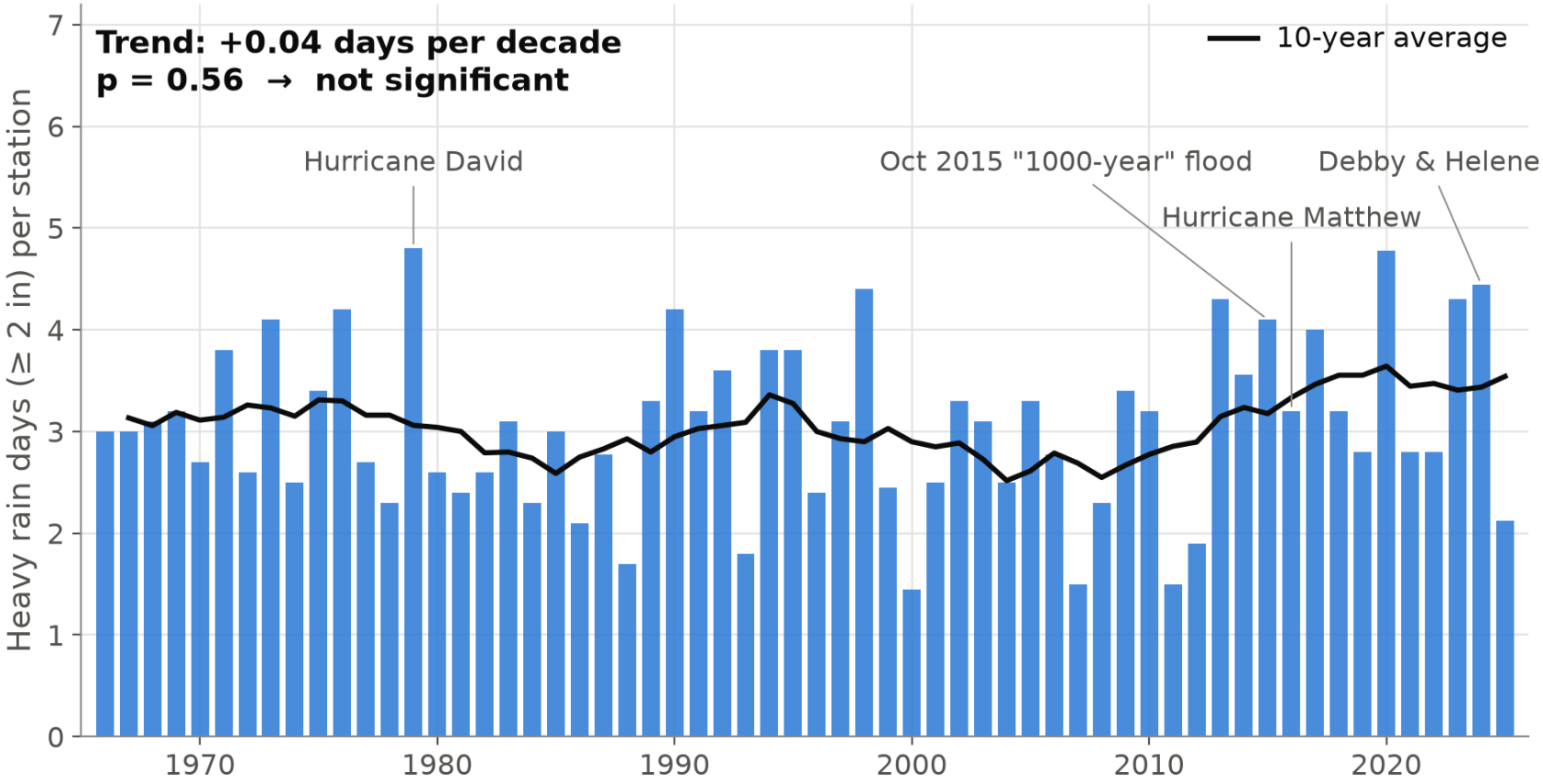
longer freeze-free (growing) season than in the 1960s

- Freeze days: 46 → 38 per year (−18%)
- −3.1 freeze days per decade ($p < 0.001$)
- Good for some crops, but also for pests and pollen

Source: NOAA NCEI GHCN-Daily, 10 South Carolina stations, 1966-2025

ANGLE 3: WHAT PEOPLE FEEL

Heavy rain: no clear trend



p = 0.56

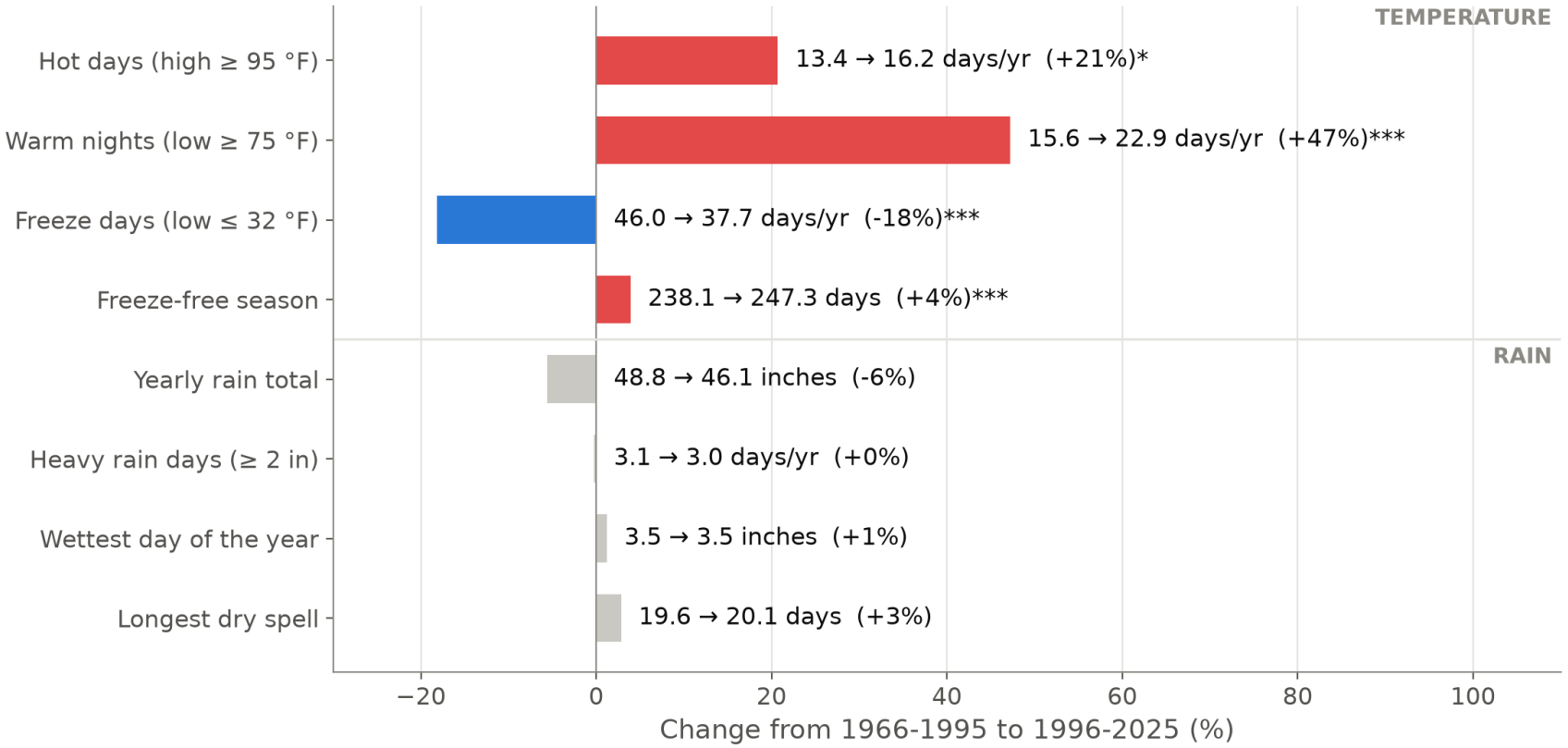
trend in heavy-rain days (≥ 2 in) is not significant

- Spikes are single storms: David, 2015 flood, Matthew, Debby & Helene
- Big storm years in every decade
- Rising since 2010, worth watching

Source: NOAA NCEI GHCN-Daily, 10 South Carolina stations, 1966-2025

ANGLE 3: WHAT PEOPLE FEEL

Summary: temperature changed, rain did not



4 of 4

temperature measures changed significantly, but 0 of 4 rain measures did

- Colored bars have a significant 60-year trend
- Gray bars could be natural ups and downs
- Rain varies more, so it needs more data

Source: NOAA NCEI GHCN-Daily, 10 South Carolina stations, 1966-2025

Conclusions

WHERE

+2.9 °F since 1966 at every trusted station. Columbia fastest (+0.58 °F/decade), tied with 3 others.

WHEN

Warming is strongest in winter and at night, and nights warm fastest in cities.

WHAT PEOPLE FEEL

+47% warm nights, −18% freezes, 3-week longer growing season. Heavy rain: no clear trend yet.

Limitations raw station data with a simple neighbor check, cities grew around some stations, a straight line is a simple model, only 10 stations

Lesson check the data first. One broken thermometer in Sumter would have changed the answer.

Thank you! Questions?

Everything is reproducible: python run_all.py rebuilds every chart in about 20 seconds

Code & data: github.com/GITGUD360/P1-data-analysis-and-visualization

Report: report/Project_Report.pdf

Data: NOAA NCEI GHCN-Daily (ncei.noaa.gov)

